

Karam Matar

Örkelljunga – Sweden

☎ (+46) 760 183 374 • ✉ karammtr@gmail.com • 🌐 karmasora.github.io/Portfolio/
in karam-matar-b92639293 • 🌀 KarmaSora

Software developer with experience in agile development, full-stack systems, CI/CD-related tooling, and low-level software. I have worked with C++, Java, Python, SQL, and Azure DevOps automation, with a strong interest in systems programming, debugging, software architecture, and building reliable developer tooling. I am currently also completing a bachelor thesis in collaboration with Iquest AB focused on an autonomous multi-agent software engineering workflow for bug fixing, verification, and pull request creation.

Professional Experience

Softhouse

Karlskrona, Sweden

Software Developer Intern

Jan 2026 – May 2026

- Worked as part of an 11-person full-stack team to develop TeamTemp, an internal employee survey platform for creating, answering, and following up on workplace surveys.
- Coordinated communication between developers, the customer, and the head of department to clarify requirements, share project updates, and support sprint planning.
- Contributed to requirements analysis, feature development, technical documentation, testing, and code reviews within an agile development process.
- Helped translate customer feedback into development tasks and supported the team in keeping the project aligned with stakeholder expectations.

Trafikverket

Remote via Blekinge Institute of Technology

Software Developer Intern

Jan 2025 – May 2025

- Collaborated with a team of six to build a tool for Trafikverket that automatically checks for vulnerabilities in third-party packages.
- Refactored a legacy PowerShell script into JavaScript to create an Azure DevOps pipeline task that generates SBOM files in JSON format.
- Worked with CI/CD-related tooling and automation in a development workflow focused on software quality and security.
- Took initiative in team communication by keeping absent team members informed and actively following up to help resolve blockers during development.

Blekinge Institute of Technology

Karlskrona, Sweden

Teaching Assistant (C++ Course)

Sep 2024 – Jan 2025

- Mentored first year students in C++ through lab supervision and one on one tutoring, focusing on core programming logic and best practices.
- Assisted students in debugging and mastering low-level concepts, including pointers, memory management, and object-oriented design.
- Guided students in analyzing program behavior and performance to ensure code correctness and efficiency.

Education

B.Sc. Software Engineering

Karlskrona, Sweden

Blekinge Institute of Technology

08/2023 – 06/2026 (Completed)

- Focus: systems programming, memory management, data structures, and building efficient software.
- Relevant Coursework: Compiler Design, Operating Systems, Algorithms & Data Structures, Linear Algebra.
- Bachelor Thesis in collaboration with Iquest AB: ACID Bot, an autonomous multi-agent AI feedback loop for bug-fix workflows, verification, and pull request creation across external code repositories.

Exchange Semester in Computer Science (Fall 2025)

Newcastle upon Tyne, United Kingdom

Newcastle University

09/2025 – 01/2026 (Completed)

- Focus: data analysis and cleaning datasets, basic machine learning algorithms, secure software development, threat analysis, and systems design for games.
- Relevant Coursework: Computer Vision & AI, Algorithm Design and Analysis, Security Programming, Game Design.

Projects

TeamTemp – Employee Survey Platform (React, TypeScript, Express, Prisma, MongoDB, Bun): Built a multi-platform employee pulse survey system for Softhouse as part of an 11-person agile team. The system includes an admin

portal, user portal, React Native/Expo mobile app, and an Express API with Prisma/MongoDB. Key functionality includes survey creation, role-based access control, Google OAuth authentication, email and Slack reminders, Qlik dashboard integration, localization in Swedish and English, and survey follow-up analytics. The project involved requirements analysis, full-stack development, testing, technical documentation, code reviews, and continuous communication with the customer and head of department.

Bachelor Thesis: ACID Bot (JS, Git, CI/CD, Multi-Agent Systems): Developing an autonomous multi-agent software engineering workflow in collaboration with Iquest AB. The system fetches bug-tagged tasks from ClickUp, analyzes and fixes issues in an external target repository through controlled Git access, verifies whether the fix is valid, updates task state, and creates pull requests for human review. The architecture includes bounded retries per agent for cost and safety control, and is designed to run on a server or inside CI/CD pipelines. This work involves workflow design, repository automation, validation logic, requirements handling, and reliable orchestration across multiple agents.

Compiler from Scratch (C++): Built an end-to-end compiler including lexer, parser, semantic analysis, CFG-based IR, TAC generation, and a custom virtual machine. This project strengthened my understanding of low-level software behavior, system design, and structured debugging.

VulnerabilityChecker2.0 (PowerShell, JavaScript, Azure DevOps): Designed and implemented an Azure DevOps pipeline task for detecting vulnerabilities in third-party dependencies across multiple repositories, parsing dependency manifests, querying vulnerability data, and generating Software Bill of Materials (SBOM) files in JSON format for CI/CD workflows.

ShowcaseSERL (Next.js, React, TypeScript, Tailwind CSS): Built a full-stack web application to showcase student and researcher projects. Features include dynamic listings, detailed project views, and a kiosk mode, with a focus on component-based architecture and clean data handling.

Death Marching Souls (C#, Unity): Developed a 3D action-platformer featuring player progression, enemy AI, boss encounters, and UI systems, implementing health management, experience-based leveling, and modular enemy behavior.

GyaGame (Java): Developed a 2D semi-open-world shooter game in Java using a provided graphics library, focusing on game logic, entity management, and player interaction.

Space Invaders (C++, SFML): Created a Space Invaders game in C++ using the SFML graphics library, implementing core game loops, collision detection, and rendering.

Skills

Languages: Swedish (Fluent), English (Fluent), Arabic (Native)

Programming: C++, Java, Python, C#, JavaScript, TypeScript

Systems: Systems programming, memory management, debugging, object-oriented design, algorithms & data structures

Frameworks: React, React Native/Expo, Next.js, SvelteKit, Express.js, Node.js, Prisma

DevOps/Data: Azure DevOps, Bitbucket Pipelines, CI/CD, Git, MySQL, MongoDB, Bun

Testing: Cypress (E2E), Bun Test, Jest

Other: HTML5, CSS, Tailwind CSS, SCSS, Unity, SFML

Driving License: AM/B

Events and Achievements

Hackathon: Hollytech Participated in a hackathon by Hollytech and won first place.